

MongoDB Database Dump — Verification and Submission Record

AI-Driven Autonomous NoSQL Query Optimization Framework

1. Project Information

Project title	AI-Driven Autonomous NoSQL Query Optimization Framework
Course	BCSE406L – NoSQL Databases
Database technology	MongoDB (NoSQL document database)
Dataset profile	STANDARD (CommerceBench dataset)

2. Dump Artifact

Filename	commerce-standard.archive.gz
Relative project path	artifacts/mongodb-dump/commerce-standard.archive.gz
Database	commerce
Source	mongo-monitored
Compressed size	Approximately 5.7 MB (5,928,416 bytes as stored)
SHA-256 checksum	57d15d28627d5871eb25decf103a0476279d4b57e36758fa295a47ae28890c90

3. Dataset Contents

The archive contains the following collections from the commerce database. Document counts are the verified collection counts.

Collection	Document Count
customers	20,000
products	10,000
orders	200,000
events	500,000
inventory	30,000
permission_probe	0
TOTAL	760,000

The dump excludes the temporary optimization-run collection r24r27_approval_success_2d572a984a.

4. Restore Verification

The submitted archive was restored into a temporary MongoDB database as an integrity check.

- Documents restored: **760,000**
- Restore failures: **0**
- Temporary restore database: removed after verification
- Original STANDARD dataset: retained intact

5. Scope and Purpose

This artifact is the MongoDB database dump supporting the Review 2 working prototype of the AI-Driven Autonomous NoSQL Query Optimization Framework. It contains the CommerceBench STANDARD dataset used to demonstrate the implemented NoSQL query optimization workflow. The project's control plane is a separate PostgreSQL database and is not part of this dump.

The project operates against MongoDB through a restricted executor permission boundary: query-settings metadata operations are permitted, while application-document insert, update, replace, and delete operations and destructive collection and database operations are denied.

6. Submission Note

The accompanying `.archive.gz` file is the actual MongoDB dump. This PDF is its human-readable verification record and is not a replacement for the binary dump.